Sijil: one employee, one truth

Matches the same employee across HR systems, finds and explains every inconsistency, suggests the likely correct value with its sources, ranks by business impact, and shows what a fix would touch before a person approves it.

14,806	322/322	0	254/254
records matched, 0 wrong	planted errors found	false positives	suggested values correct

Data limitation. The official data pack and test set were not published, so we curated a synthetic sample in the brief's shape (four HR systems, about 5,000 people, seed 2310, never used while building). Results show the approach on that sample, not on Emirates data. **All findings and files:** github.com/pratikkp24/aonami-sijil-emirates-hr-data-quality

Try the live prototype
aonami-sijil.vercel.app
Hosted, no install. Press **Guided demo**.
Demo video www.loom.com/share/...

THE APPROACH

Sijil treats core HR as the spine and joins every other system to it with **deterministic, explainable matching**. It checks each matched person against the data dictionary and the other systems, then uses the dependencies file to rank each issue by what it breaks: payroll, the WPS file, the nationalisation report, leadership headcount. **No cloud AI is used**: the same pack always gives the same issues, and every suggested value shows the sources and the rule behind it.



DETECTION AND MATCHING ACCURACY

Before the official pack, we built our own in the brief's shape: 5,000 people held by core HR, payroll (different IDs, "LAST, First", DD/MM/YYYY), learning (joined by email, department names) and the directory (with leavers, contractors and service accounts), with **322 planted errors** across 14 types and **48 near misses** that must not be flagged: transliterated names, a missing payroll reference, a move inside the learning sync window, someone who left today, service accounts, and namesakes with the same birth date.

BENCHMARK	RESULT
Records matched to the right person	14,805 of 14,805
Planted errors found (recall)	322 of 322
False positives (precision)	0 (100%)
Suggested value correct	254 of 254
Four more generated packs	Same results
Renamed columns, other date formats	Identical issues
Run time, 4 systems, in a browser	about 40 ms

EXPLANATION AND PRIORITISATION

Each issue reads in plain language, with the value in every system, the suggested value and why:

```
P1 · Leaver still paid · PAYROLL pay_status
Mariam Qureshi left on 6 Jan 2026 but payroll is still
active and paid them on 27 Sep 2026: about AED 30,653
a month to a leaver.
CORE: Terminated 2026-01-06 · PAYROLL: ACTIVE, last
paid 2026-09-27
Suggested: STOPPED (high confidence). Core HR is the
system of record for status; left 279 days ago.
Touches: Monthly payroll run (weight 5), WPS salary
file (weight 5)
```

Ranking starts from what the error risks (money paid in error, access by leavers) and adds the weight of every report that uses the field, from the dependencies file. A duplicate only distorts core reports; a ghost on payroll hits the payroll run and the WPS file, so it ranks higher. **The impact view** follows the feeds: fixing in core HR lets the daily feeds correct payroll, learning and the directory; fixing only downstream would be overwritten.

Trust scores per field and per report show which numbers can go to leadership and the regulator, and the question box ("How many active employees in Dubai?") answers with its source, the filters used, the field's trust, and the payroll count beside it.

RESPONSIBLE AI AND SECURITY STATEMENT

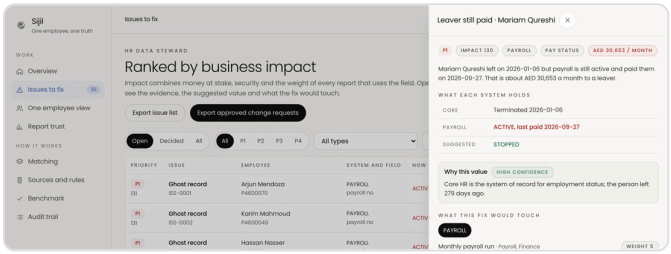
Sijil suggests; a person approves. Nothing is written back to any HR system. Every fix needs a named person to approve, reject or defer it, with the value they chose and a note; approved fixes are exported as change requests for the system owner. Every load and decision is in the audit trail.

No AI model touches the data. Matching and checks are deterministic rules that can be read and tested; every suggested value shows its sources and logic, and ambiguous matches are never forced. The synthetic pack was processed only on our own machine and was not uploaded to any service that keeps or trains on data. The hosted demo runs on Aonami's own synthetic sample; files loaded there stay in the viewer's browser.

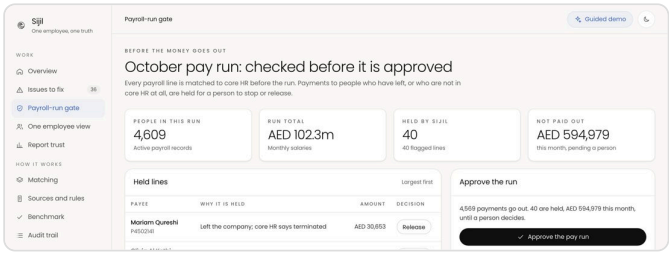
- **Data protection:** designed for the UAE Personal Data Protection Law (Federal Decree-Law 45 of 2021): data stays in Emirates' environment, the tool only reads, and outputs carry record IDs. It follows the principles of the UAE Charter for the Development and Use of AI: transparency, human oversight and accountability.
- **Limits we disclose:** results above are on our own synthetic benchmark, where we also wrote the data; the official labelled pack is the real test. Systems of record and sync windows come from the data dictionary and should be confirmed with each system owner.

HOW IT RUNS INSIDE EMIRATES' ENVIRONMENT

- **Deployment:** a static web application and a command-line runner, packaged as a container in Emirates' private cloud or on premises. No outbound connection and no cloud AI.
- **Sources:** read-only extracts or API pulls from each HR system. Adding a source is configuration: a column profile and the system's authority per field in the data dictionary. The 50+ HR applications can be added one at a time.
- **Access:** Emirates single sign-on with roles: data steward, analytics lead, systems owner. Change requests go to the ticketing tool each system owner already uses.
- **Schedule:** the runner checks nightly; new and resolved issues are tracked over time, so the trust scores show whether data quality is improving.



A leaver, still paid. Every system's value, the suggested value with its logic, and the reports and systems the fix would touch, before a person approves.



The payroll-run gate. Payroll lines are matched to core HR before the monthly run, so payments to leavers are held before the money leaves.

PILOT NEXT QUARTER

WEEKS	ACTIVITY
1 to 2	Connect core HR, payroll and the directory; confirm systems of record and report weights
3 to 4	First full run; data stewards review the top 200 issues; tune rules
5 to 8	Weekly runs; approved fixes to system owners; add learning and two more systems
End	Issues fixed, AED recovered, report trust before and after; go or no-go

TEAM

Aonami builds AI agents and decision systems for regulated institutions, with a named person approving every action. A team of nine across engineering, AI and delivery; co-founders Pratik Kumar and Jayesh Kumar Rout, with backgrounds in lending, fintech and risk at Stashfin, OneCard and Northern Trust.

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